

# Hydra Model Results Summary

## NWM Streamflow Error Correction — Week 2 Progress Report

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## 1 Executive Summary

This report presents results from the Hydra model development for post-processing National Water Model (NWM) streamflow predictions. We compare three architectures across four study sites in the Upper New River / Watauga River watersheds:

- **LSTM Baseline:** Standard recurrent approach
- **Hydra v1:** Transformer-only architecture
- **Hydra v2:** GRU  $\rightarrow$  Transformer hybrid (current best)

**Key Finding:** The GRU pre-encoder is essential. Hydra v2 reduces RMSE by 36–54% over v1 and achieves positive NSE at all sites where v1 fails catastrophically.

## 2 Study Sites

| Site ID  | Name                                    | Watershed | Type                | Drainage Area |
|----------|-----------------------------------------|-----------|---------------------|---------------|
| 03161000 | South Fork New River near Jefferson, NC | New River | Mid-basin           | —             |
| 03164000 | New River near Galax, VA                | New River | Mainstem integrator | —             |
| 03479000 | Watauga River near Sugar Grove, NC      | Watauga   | Headwaters          | —             |
| 03486000 | Watauga River at Elizabethton, TN       | Watauga   | Regulated (TVA)     | —             |

**Data Period:** 2010-01-01 to 2020-12-31 (hourly)

## 3 Model Architectures

### 3.1 LSTM Baseline

- 2-layer LSTM encoder (hidden=256)
- Input: 168-hour (7-day) lookback window
- Static feature fusion via concatenation
- Predicts residual:  $\hat{e}(t) = Q_{\text{obs}}(t) - Q_{\text{nwm}}(t)$

### 3.2 Hydra v1 (Transformer-Only)

- Patch convolution OR linear input projection
- Transformer encoder with **pre-norm** and **causal masking**
- Dilated convolutions (exponential: 1, 2, 4, 8)
- Channel attention (squeeze-excitation)
- Cyclic positional encoding (hour/day overlays)
- GLU fusion layer

### 3.3 Hydra v2 (GRU $\rightarrow$ Transformer Hybrid)

- **Single-layer GRU pre-encoder** (projects to  $d_{\text{model}} = 64$ )
- Transformer encoder with **post-norm**, no causal mask
- Dilated convolutions (fixed: 1, 3, 6)
- Attention pooling with learnable query
- Static feature cross-attention
- **Heteroscedastic uncertainty heads** (Gaussian NLL)

## 4 Results: GRU Ablation (v2 vs. v1)

### 4.1 RMSE Comparison ( $\text{m}^3/\text{s}$ )

| Site     | Hydra v1 (Transformer) | Hydra v2 (GRU+Transformer) | $\Delta$ Improvement |
|----------|------------------------|----------------------------|----------------------|
| 03161000 | 21.69                  | 12.33                      | 43% $\downarrow$     |
| 03164000 | 97.21                  | 62.11                      | 36% $\downarrow$     |
| 03479000 | 11.98                  | 7.37                       | 38% $\downarrow$     |
| 03486000 | 49.45                  | 22.55                      | 54% $\downarrow$     |

### 4.2 NSE Comparison

| Site     | Hydra v1 | Hydra v2 | $\Delta$ Improvement |
|----------|----------|----------|----------------------|
| 03161000 | -0.35    | 0.56     | +0.91                |
| 03164000 | -0.62    | 0.34     | +0.96                |
| 03479000 | 0.05     | 0.64     | +0.59                |
| 03486000 | -0.88    | 0.61     | +1.49                |

| Site     | Hydra v1 | Hydra v2 | $\Delta$ Improvement |
|----------|----------|----------|----------------------|
| 03161000 | -69.1    | -16.9    | +52 pts              |
| 03164000 | -81.3    | -2.8     | +78 pts              |
| 03479000 | -59.0    | 4.7      | +64 pts              |
| 03486000 | -72.2    | -4.3     | +68 pts              |

  

| Site     | Hydra v1 | Hydra v2 | $\Delta$ Improvement |
|----------|----------|----------|----------------------|
| 03161000 | 0.494    | 0.781    | +0.29                |
| 03164000 | -0.039   | 0.599    | +0.64                |
| 03479000 | 0.667    | 0.804    | +0.14                |
| 03486000 | 0.649    | 0.784    | +0.14                |

### 4.3 PBIAS Comparison (%)

### 4.4 Correlation Coefficient (CC)

## 5 Results: Hydra v2 vs. NWM Baseline

| Site            | Metric | NWM Baseline | Hydra v2 | Improved? |
|-----------------|--------|--------------|----------|-----------|
| <b>03161000</b> | RMSE   | 12.36        | 12.33    | ✓ (0.2%)  |
|                 | NSE    | 0.562        | 0.564    | ✓         |
|                 | PBIAS  | -13.2%       | -16.9%   | ✗         |
|                 | CC     | 0.763        | 0.781    | ✓         |
| <b>03164000</b> | RMSE   | 63.41        | 62.11    | ✓ (2.0%)  |
|                 | NSE    | 0.312        | 0.339    | ✓         |
|                 | PBIAS  | -9.5%        | -2.8%    | ✓         |
|                 | CC     | 0.594        | 0.599    | ✓         |
| <b>03479000</b> | RMSE   | 7.49         | 7.37     | ✓ (1.6%)  |
|                 | NSE    | 0.630        | 0.642    | ✓         |
|                 | PBIAS  | +7.1%        | +4.7%    | ✓         |
|                 | CC     | 0.802        | 0.804    | ✓         |
| <b>03486000</b> | RMSE   | 27.15        | 22.55    | ✓ (17.0%) |
|                 | NSE    | 0.433        | 0.609    | ✓ (+0.18) |
|                 | PBIAS  | -9.8%        | -4.3%    | ✓         |
|                 | CC     | 0.674        | 0.784    | ✓ (+0.11) |

**Summary:** 3 of 4 sites show improvement on all metrics. Site 03161000 shows marginal gains but PBIAS worsens.

## 6 Why Does the GRU Component Help?

The GRU pre-encoder provides three critical benefits for streamflow time series:

### 6.1 Sequential Inductive Bias

Transformers treat input tokens as an unordered set unless positional encoding is strong. Streamflow is inherently sequential—discharge at hour  $t$  depends causally on hours  $t - 1$ ,  $t - 2$ , etc. The GRU

**enforces temporal ordering** through its recurrent structure, providing a stronger inductive bias than positional encoding alone.

## 6.2 Smooth Hidden State Transitions

GRUs produce **continuous, smooth representations** across time via their gating mechanism. This is well-suited to streamflow, which exhibits gradual rises and recessions. Pure Transformers can produce discontinuous attention patterns that struggle with the smooth dynamics of hydrological processes.

## 6.3 Compression Before Attention

The GRU acts as a **temporal compressor**, extracting local patterns before the Transformer’s global attention. This reduces the burden on the Transformer to learn both local and global dependencies simultaneously. The Transformer can focus on longer-range relationships (e.g., lag between precipitation and discharge) while the GRU handles short-term autocorrelation.

## 6.4 Gradient Stability

The GRU’s gating provides a stable gradient pathway during training. Transformers can suffer from attention collapse or instability on small datasets. The GRU acts as a regularizer, stabilizing early training dynamics.

## 6.5 Empirical Evidence

- v1 (Transformer-only) achieves **negative NSE** on 3/4 sites (worse than predicting the mean)
- v1 shows massive negative bias (PBIAS -59% to -81%), indicating systematic underprediction
- v2 (GRU+Transformer) stabilizes all metrics to reasonable ranges

This suggests v1 fails to learn meaningful temporal representations from position encoding alone.

# 7 Problem Area: Site 03161000

Site 03161000 (South Fork New River) shows:

- Marginal RMSE/NSE improvement ( $< 1\%$ )
- **PBIAS worsens** from -13.2%  $\rightarrow$  -16.9%

## 7.1 Possible Causes

1. **Mid-basin complexity:** Receives contributions from multiple tributaries with different lag times
2. **Limited improvement ceiling:** NWM baseline already performs reasonably well (NSE=0.56)
3. **Loss function:** MSE-based loss doesn’t penalize systematic bias

## 7.2 Proposed Fixes

1. Add PBIAS penalty to loss function
2. Reintroduce cyclic time encoding (present in v1, removed in v2)
3. Post-hoc bias calibration (partially attempted, needs refinement)

## 8 Architecture Improvement Proposals

| Modification                       | Expected Impact                    | Implementation Effort |
|------------------------------------|------------------------------------|-----------------------|
| Cyclic time encoding (hour/day)    | Better seasonality, may help PBIAS | Low (copy from v1)    |
| Causal masking in Transformer      | Stricter temporal validity         | Low                   |
| PBIAS-aware loss term              | Directly targets bias              | Medium                |
| Channel attention (squeeze-excite) | Better feature selection           | Medium                |

## 9 Conclusions

1. **The GRU component is essential** — removing it causes model failure
2. **Hydra v2 improves or matches NWM baseline** on 15/16 site-metric combinations
3. **Best results at 03486000** (regulated site): RMSE  $\downarrow$  17%, NSE +0.18
4. **One problem case:** 03161000 PBIAS worsens slightly
5. **Results are publishable** with appropriate framing of the 03161000 limitation

## 10 Recommended Next Steps

### 10.1 If results are acceptable:

- Proceed to WRR manuscript draft
- Generate publication figures (map, bar charts, hydrographs)
- Frame 03161000 limitation in Discussion section

### 10.2 If improvement needed:

- 1-day timebox: cyclic encoding + PBIAS loss for 03161000
- Re-run only the problem site
- Do not delay manuscript start beyond Feb 5

## 11 Appendix: Full Metrics Table

| Site     | Model        | RMSE  | NSE    | PBIAS (%) | CC     |
|----------|--------------|-------|--------|-----------|--------|
| 03161000 | NWM Baseline | 12.36 | 0.562  | -13.2     | 0.763  |
| 03161000 | Hydra v1     | 21.69 | -0.350 | -69.1     | 0.494  |
| 03161000 | Hydra v2     | 12.33 | 0.564  | -16.9     | 0.781  |
| 03164000 | NWM Baseline | 63.41 | 0.312  | -9.5      | 0.594  |
| 03164000 | Hydra v1     | 97.21 | -0.620 | -81.3     | -0.039 |
| 03164000 | Hydra v2     | 62.11 | 0.339  | -2.8      | 0.599  |
| 03479000 | NWM Baseline | 7.49  | 0.630  | +7.1      | 0.802  |
| 03479000 | Hydra v1     | 11.98 | 0.055  | -59.0     | 0.667  |
| 03479000 | Hydra v2     | 7.37  | 0.642  | +4.7      | 0.804  |
| 03486000 | NWM Baseline | 27.15 | 0.433  | -9.8      | 0.674  |
| 03486000 | Hydra v1     | 49.45 | -0.880 | -72.2     | 0.649  |
| 03486000 | Hydra v2     | 22.55 | 0.609  | -4.3      | 0.784  |